

1.GENERAL
ALL ELECTRICAL WORKS SHALL BE IN ACCORDANCE:

- LATEST OPTUS D&C SPECIFICATIONS
- AS/NZ 3000:2018, AS/NZ 3008, AS/NZ 1768, AS/NZ3015 (LATEST EDITIONS AND AMENDMENTS AT TIME OF CONSTRUCTION SHALL APPLY.
- TAS NETWORKS SERVICE AND INSTALLATION RULES

CONTRACTOR TO ENSURE FAULT CURRENT RATINGS OF PROTECTIVE EQUIPMENT INSTALLED ARE ADEQUATE FOR PRSPECTIVE FAULT LEVELS.

CABLE ROUTES SHALL BE AS DIRECT AS POSSIBLE. WHERE CABLE TURNS ARE REQUIRED, THE RADIUS OF ANY BENDS SHALL NOT BE LESS THAN THE MINIMUM BENDING RADIUS OF THE CABLE.

ALL EXPOSED CABLE, CONDUITS AND NON-METALLIC FASTENERS SHALL BE UV RESISTANT AND SHALL BE ADEQUATELY SECURED TO SURFACE, CABLE TRAY OR SRTUCTURE.

CONTRACTOR SHALL MAKE THEMSELVES AWARE OF ALL SERVICES PRESENT ON SITE AND OF ALL SITE CONDITIONS AND SAFETY REQUIREMENTS PRIOR TO COMMENCING WORK ON SITE.

2.OPTUS SUPPLY OF ELECTRICITY
EXISTING OPTUS AC POWER CAPACITY ON THE SITE IS 63A SINGLE PHASE 415V AND IT REQUIRES TO BE UPGRADED TO 150A 1-PHASE AS PART OF PROPOSED 5G AUGMENTAITON WORKS.

150A 1- PHASE SUPPLY SERVICE FOR OPTUS SHALL BE TAKEN FROM THE NEW UPGARDED ELECTRICAL MAIN SWITCH BOARD TO BE INSTALLED INSIDE THE TELCO COMPOUND AS SHOWN ON SHEET E2 AND G2.

UPGRADED SITE ELECTRICAL MAIN SWITCH BOARD AC POWER WILL BE SUPPLIED FROM THE EXISTING POWER AUTHORITY POWER POLE No 399256 AS SHOWN ON SHEET G2 AND E2.

3. DISTRIBUTION MAINS CABLE
CURRENTLY ON THE SITE 2 x 1c x 25mm2 Al/ABC/XLPE OVERHEAD LOW VOLTAGE DISTRIBUTION MAINS CABLE IS INSTALLED BETWEEN THE EXISTING POLE MOUNTED 50kVA SINGLE PHASE TRANSFORMER No T 62 1137 AND THE POWER AUTHORITY POLE No 399256 AS SHOWN ON SHEET E2 AND G2.

EXISTING OVERHEAD LOW VOLTAGE DISTRIBUTION MAINS CABLE TO BE UPGRADED BY THE POWER AUTHORITY. THIS SECTION WILL BE UPDATED ONCE POWER APPROVAL IS RECIVED FROM THE POWER AUTHORITY.

4. CONSUMERS MAINS CABLE
CURRENLTY 2 x 1c x 16mm² Cu/XLPE UGOH UNMETERED OPTUS CONSUMERS MAINS CABLE IS INSTALLED BETWEEN THE EXISTING POWER AUTHORITY POLE No 399256 AND THE EXISTING OPTUS SHELTER EXTERNAL MOUNTED METER PANEL.

CURRENLTY 2 x 1c x 16mm2 Cu/XLPE UGOH UNMETERED NBN CONSUMERS MAINS CABLE IS INSTALLED BETWEEN THE EXISTING POWER AUTHORITY POLE No 399256 AND THE EXISTING NBN H-FRAME MOUNTED METER PANEL.

ELECTRICIAN ON THE SITE SHALL DISCONNECT AND RECOVER BOTH OPTUS AND NBN INSTALLED CONSUMERS MAINS CABLE.

ELECTRICIAN ON THE SITE SHALL PROVIDE AND INSTALL NEW 2 x 1c x 95mm2 Cu/XLPE/PVC UGOH ON THE EXISTING POLE No 399256 WITHIN NEW 63mm HD uPVC CONDUIT AND SHALL TERMINATE INTO NEW TELCO COMMON ELECTRICAL MAIN SWITCH BOARD AS SHONW ON SHEET E2.

5. MAIN SWITCH BOARD
ELECTRICIAN ON THE SITE SHALL INSTALL NEW MINIMUM 300A RATED CONCRETE PLINTH MOUNTED ELECTRICAL MAIN SWITCH BOARD AS SHOWN ON SHEET G2 AND E2.

ELECTRICIAN ON THE SITE SHALL PROVIDE AND INSTALL BELOW LISTED NEW ELECTRICAL SWITCH GEARS WITHIN NEW TELCO ELECTRICAL MAIN SWITCH BOARD:

- > NEW SUITABLE SIZE BUSBARS OR ACTIVE/NEUTRAL LINKS TO TERMINATE 95mm2 CONSUMERS MAINS CABLE
- > NEW 250A 1-POLE MCCB SERVICE PROECTION DEVICE WITH TRIP CURRENT SETTING SET AT 230A
- > NEW UNMETERED ACTIVE AND NEUTRAL LINKS
- > NEW 150A 1-POLE FOR OPTUS AS METER PROTECTION DEVICE
- > NEW SINGLE PHASE CURRENT TRANSFORMER METERING EQUIPMENT FOR OPTUS
- > NEW 150A 1-POLE MCB C-CURVE FOR OPTUS
- > RELOCATE EXISTING NBN SERVICE FUSE FROM ITS METER PANEL TO NEW ELECTRICAL MAIN SWITCH BOARD
- > RELOCATED EXISTING NBN SINGLE PHASE METER
- > INSTALL NEW 80A 1-POLE MCB FOR NBN
- > NEW MEN LINKS AND EARTH ELECTRODE

ELECTRICAIN SHALL DECOMMISSION EXISTING OPTUS SINGLE PHASE METER FROM THE EXISTING METER PANEL. ALL EXISTING MULTIPLE MEN CONNECTIONS ONTHE SITE SHALL BE RECOVERED.

** NOTE - TEMPORARY SUITABLE SIZE GENERATORS WILL BE REQUIRED FOR THE EXISTING OPTUS SHELTER AND NBN UNTILL THE ELCTRICAL UPGRADE WORKS ARE COMPLETED. ELECTRICIAN SHALL NOTIFY TO NBN/OPTUS ABOUT THE PROPOSED AC UPGRADE WORKS BEFORE STARTING ANY WORKS ON THE SITE.
ELECTRICIAN SHALL GET NEW ELECTRICAL MAIN SWITCH BOARD DRAWINGS FROM THE MSB MANUFACTURER AND SHALL SUBMIT TO THE POWER AUTHORITY INSPECTOR FOR AN APPROVAL BEFORE MANUFACTURING THE NEW MSB.

6. SUBMAINS CABLE
EXISTING NBN UNDERTGROUND SUBMAINS CABLES SHALL BE FULLY RECOVERED.

ELECTRICAIN SHALL PROVIDE AND INSTALL NEW 1 x 2c x 35mm2 Cu/XLPE + E MULTICORE SUBMAINS CABLE WITHIN NEW 63mm HD CONDUIT FOR OPTUS BETWEEN THE NEW ELECTRICAL MAIN SWITCH BOARD AND OPTUS SHELTER MOUNTED EXTERNAL DISTRIBUTION BOARD AS SHOWN ON SHEET E2.

ELECTRICAIN SHALL PROVIDE AND INSTALL NEW 1 x 2c x 16mm2 Cu/XLPE + E MULTICORE SUBMAINS CABLE WITHIN NEW 63mm HD CONDUIT FOR NBN BETWEEN THE NEW ELECTRICAL MAIN SWITCH BOARD AND EXISTING NBN H-FRAME MOUNTED EXTERNAL DISTRIBUTION BOARD AS SHOWN ON SHEET E2.

7.APPLICATION FOR CONNECTION
AN APPLICATION FOR CONNECTION TO THE POWER AUTHORITY IS SUBMITTED AND REFERENCE NUMBER IS CN25-154601. THIS SECTION WILL BE UPDATED ONCE SOME REPOSENSE IS RECIEVED BACK FROM THE POWER AUTHORITY.

8.ISOLATOR SWITCH
ELECTRICAN SHALL UPGRADE THE EXISTING OPTUS ISOLATOR SWITCH INSTALLED WITHIN OPTUS SHELTER EXTERNAL AC DB WITH NEW 150A 1-POLE ISOLATOR SWITCH AS SHOWN ON SHEET E2.

9.LABELING
FIX STANDARD LABELS TO ALL FUSE, METER AND SWITCH POSITIONS IN METER ENCLOSURE LABELS TO BE BLACK LETTERING ON WHITE BACKGROUND "TRAFFOLYTE". LETTERING TO BE MINIMUM 10mm HIGH. TEXT TO READ "OPTUS" , "NBN". PROVIDE A PERMANET SKETCH PLAN IN THE NEW GROUP METERING ENCLOSURE AND OPTUS EXTERNAL SHELTER MOUNTED AC DB INDICATING THE SIZE AND DIRECTION OF THE SERVICES.

10.EARTHING
RRU'S and AAU's SHALL BE INDIVIDUALLY EARTHED AS PER OPTUS AND MANUFACTURER'S SPECIFICATIONS. UTILISE EXISTING EARTHING POINTS ON MOUNTING ARRANGEMENT IF CAPACITY EXISTS OTHERWISE PROVIDE AND INSTALL NEW EARTH PLATES. ALL EARTHING CONNECTIONS TO BE VIA SINGLE HOLE LUG AS PER OPTUS GUIDELINES.

11.GENERATOR SUPPLY
GENERATOR INLET SOCKET IS LOCATED ON OPTUS SHELTER. REFEER TO OSD-620 FOR MANUAL CHANGEOVER SWITCH DETAILS. CONTRACTOR IS TO ENSURE THAT ALL EQUIPMENT AND WIRING ARE INSTALLED AND CONFIGURED COMPLIANT WITH AS/NZ 3010 FOR ALTERNATE GENERATOR SUPPLY.